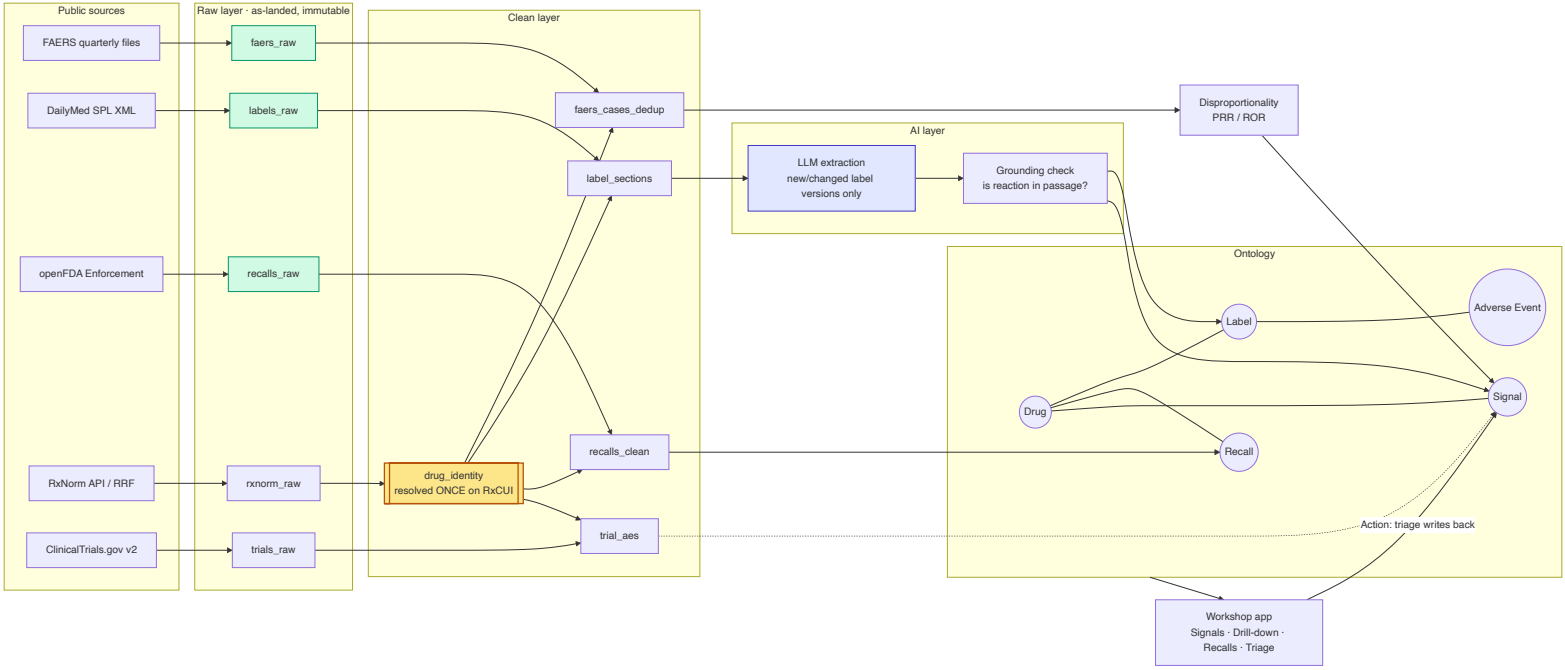
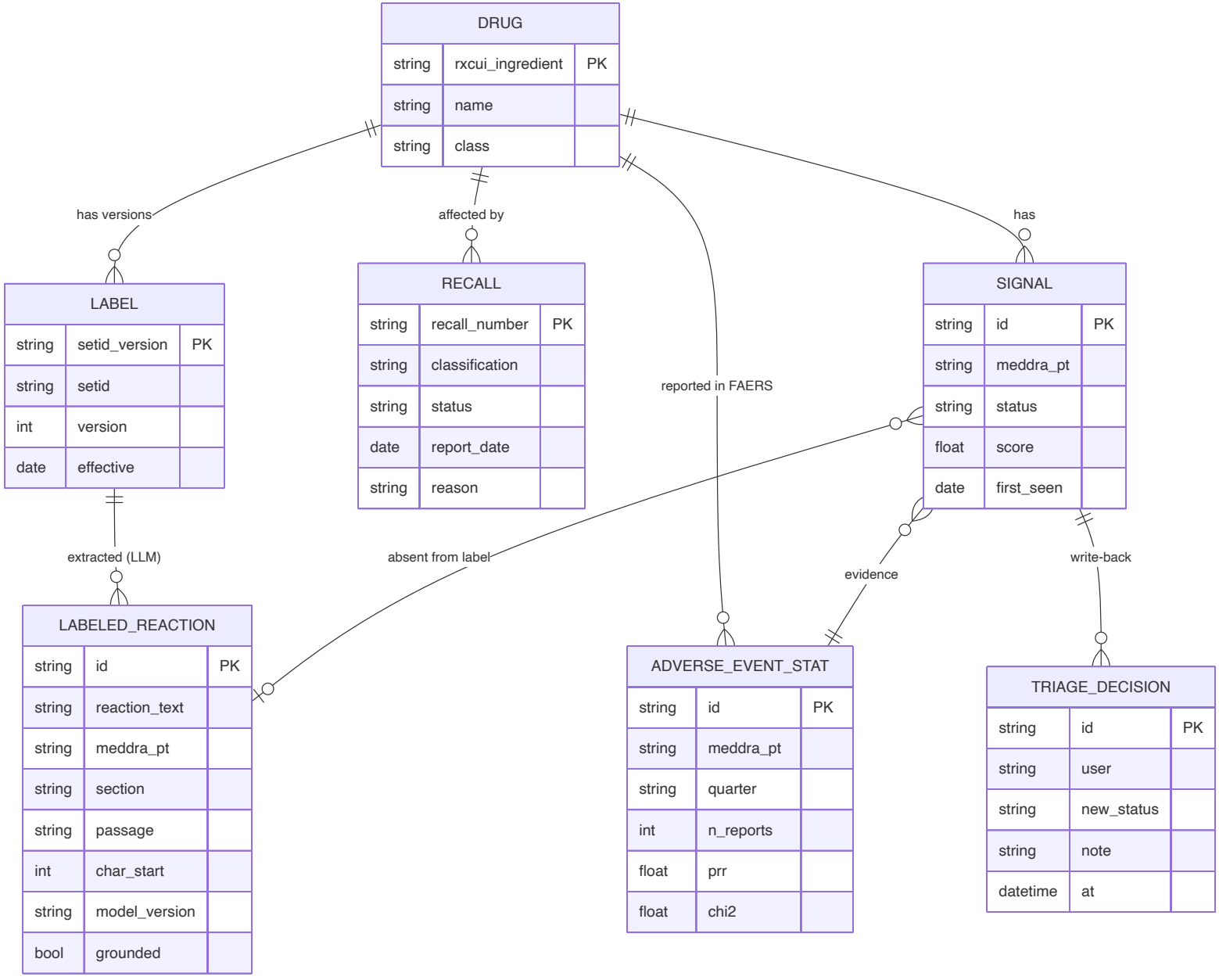
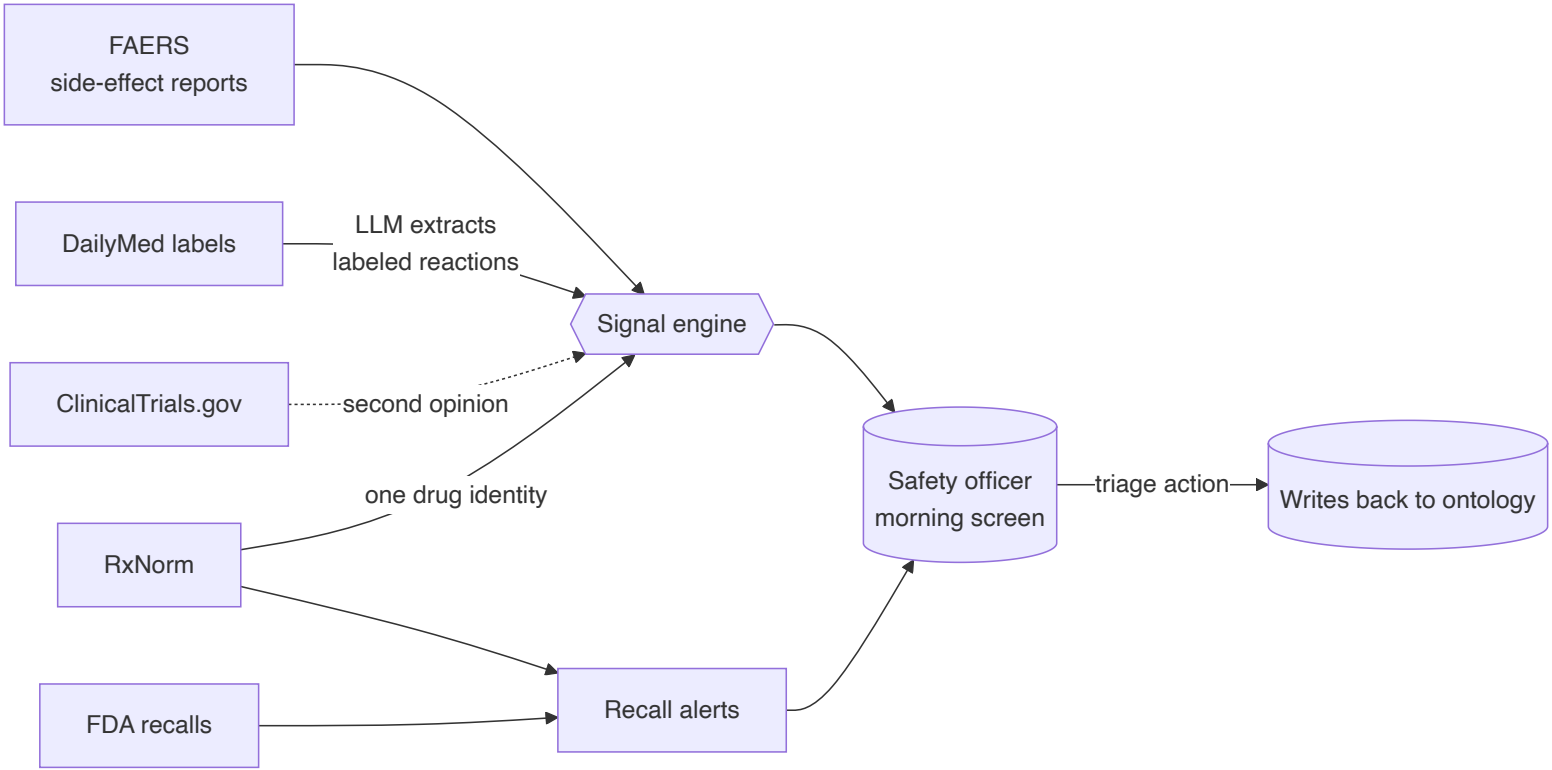
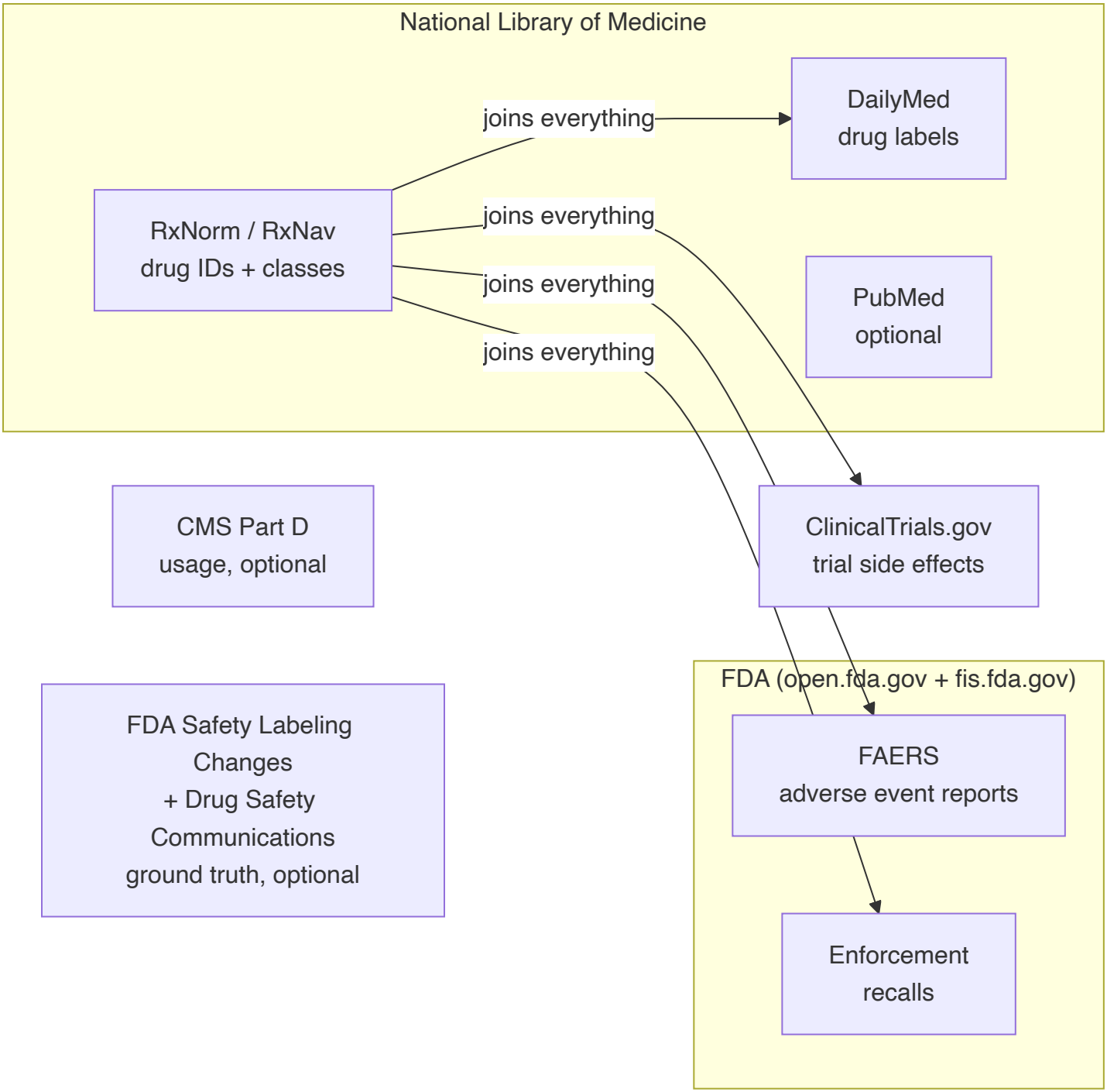
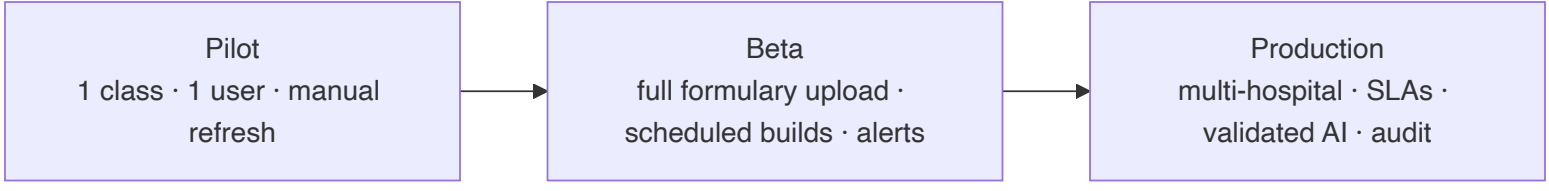

0 · Discover
Empathize
1 · Define
Scope + source profiling
2 · Pilot build
Pipelines → Ontology → AI
→ App
3 · Harden + Prove
Eval, optimization log,
monitoring
4 · Demo + Note
20-min demo, 1-page
architecture
5 · Scale path
Production roadmap


Public sources
FAERS quarterly files
DailyMed SPL XML
openFDA Enforcement
RxNorm API / RRF
ClinicalTrials.gov v2
Raw layer · as-landed, immutable
faers_raw
labels_raw
recalls_raw
rxnorm_raw
trials_raw
Clean layer
faers_cases_dedup
label_sections
recalls_clean
drug_identity
resolved ONCE on RxCUI
trial_aes
AI layer
LLM extraction
new/changed label
versions only
Grounding check
is reaction in passage?
Disproportionality
PRR / ROR
Ontology
Label
Adverse Event
Drug
Signal
Recall
Action: triage writes back
Workshop app
Signals · Drill-down ·
Recalls · Triage

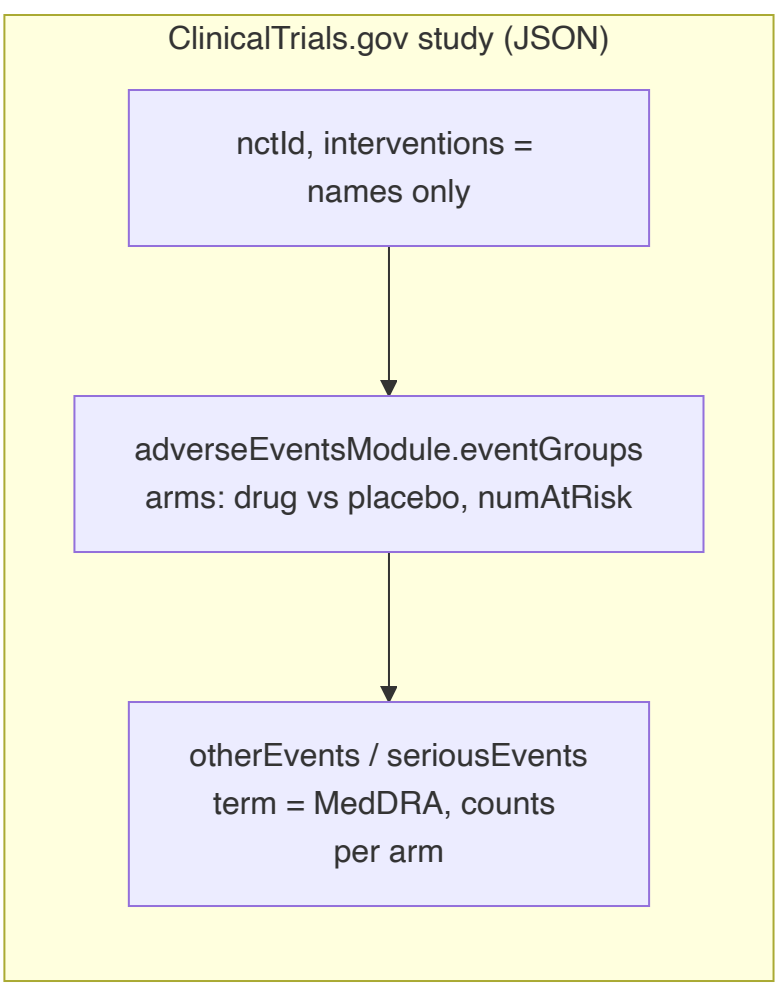

DRUG
string rxcui_ingredient PK
string name
string class
has versions
affected by
has
reported in FAERS
LABEL
string setid_version PK
string setid
int version
date effective
RECALL
string recall_number PK
string classification
string status
date report_date
string reason
SIGNAL
string id PK
string meddra_pt
string status
float score
date first_seen
extracted (LLM)
absent from label
evidence
write-back
LABELED_REACTION
string id PK
string reaction_text
string meddra_pt
string section
string passage
int char_start
string model_version
bool grounded
ADVERSE_EVENT_STAT
string id PK
string meddra_pt
string quarter
int n_reports
float prr
float chi2
TRIAGE_DECISION
string id PK
string user
string new_status
string note
datetime at

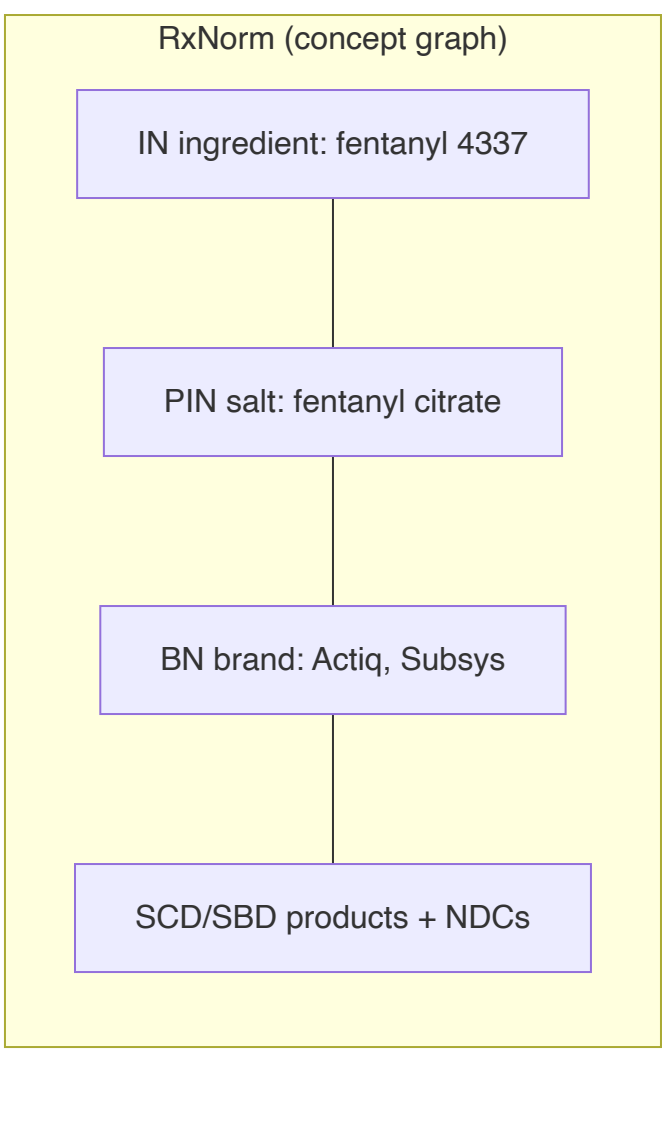

FAERS
side-effect reports
DailyMed labels
LLM extracts
labeled reactions
ClinicalTrials.gov
second opinion
one drug identity
RxNorm
FDA recalls
Signal engine
Recall alerts
Safety officer
morning screen
triage action
Writes back to ontology

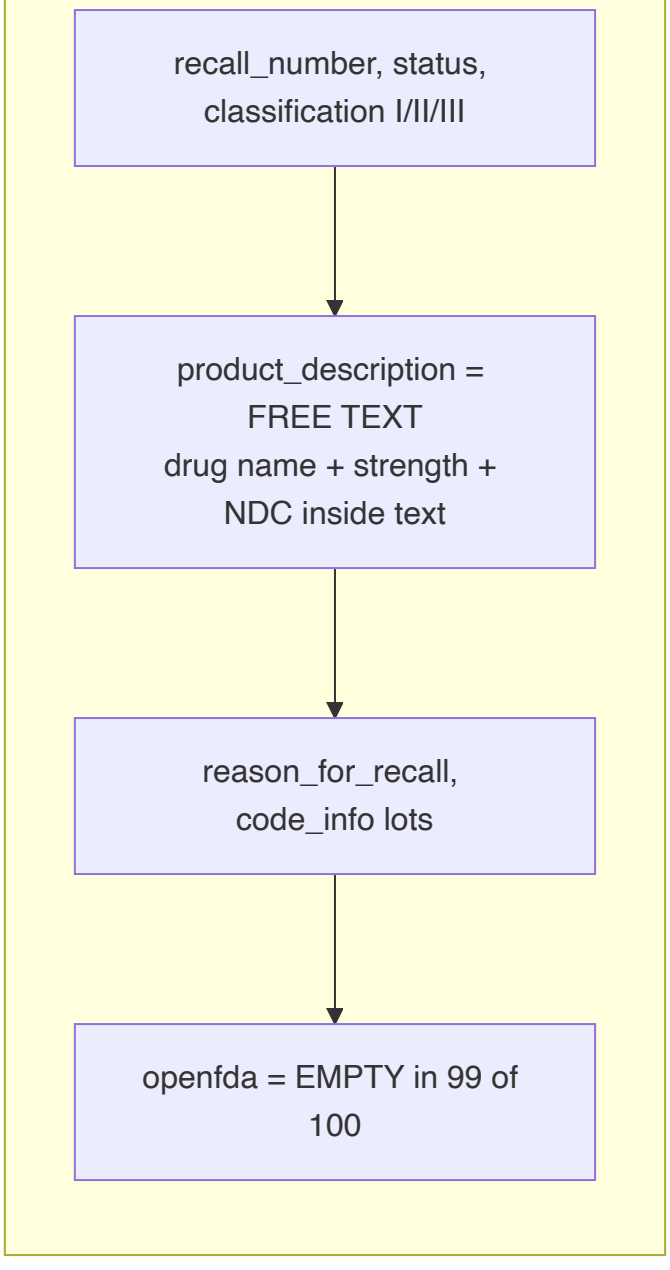

National Library of Medicine
DailyMed
drug labels
joins everything
RxNorm / RxNav
drug IDs + classes
joins everything
joins everything
joins everything
PubMed
optional
CMS Part D
usage, optional
ClinicalTrials.gov
trial side effects
FDA (open.fda.gov + fis.fda.gov)
FAERS
adverse event reports
FDA Safety Labeling
Changes
+ Drug Safety
Communications
ground truth, optional
Enforcement
recalls

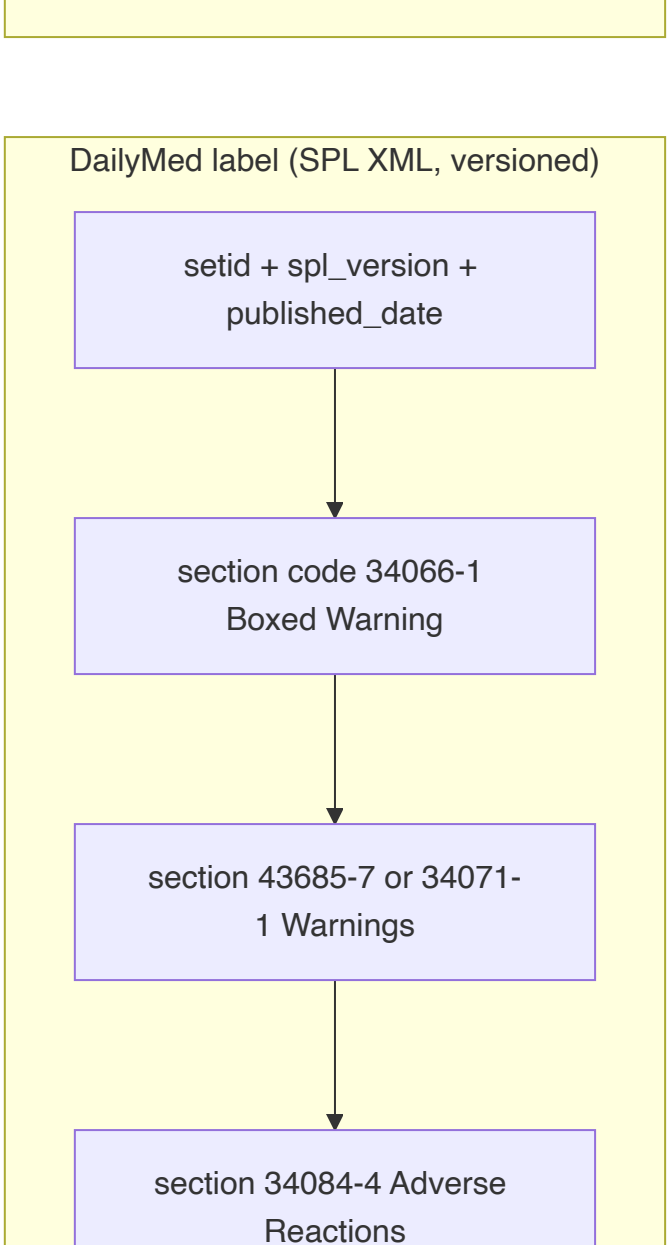

Pilot
1 class · 1 user · manual refresh
Beta
full formulary upload · scheduled builds · alerts
Production
multi-hospital · SLAs · validated AI · audit

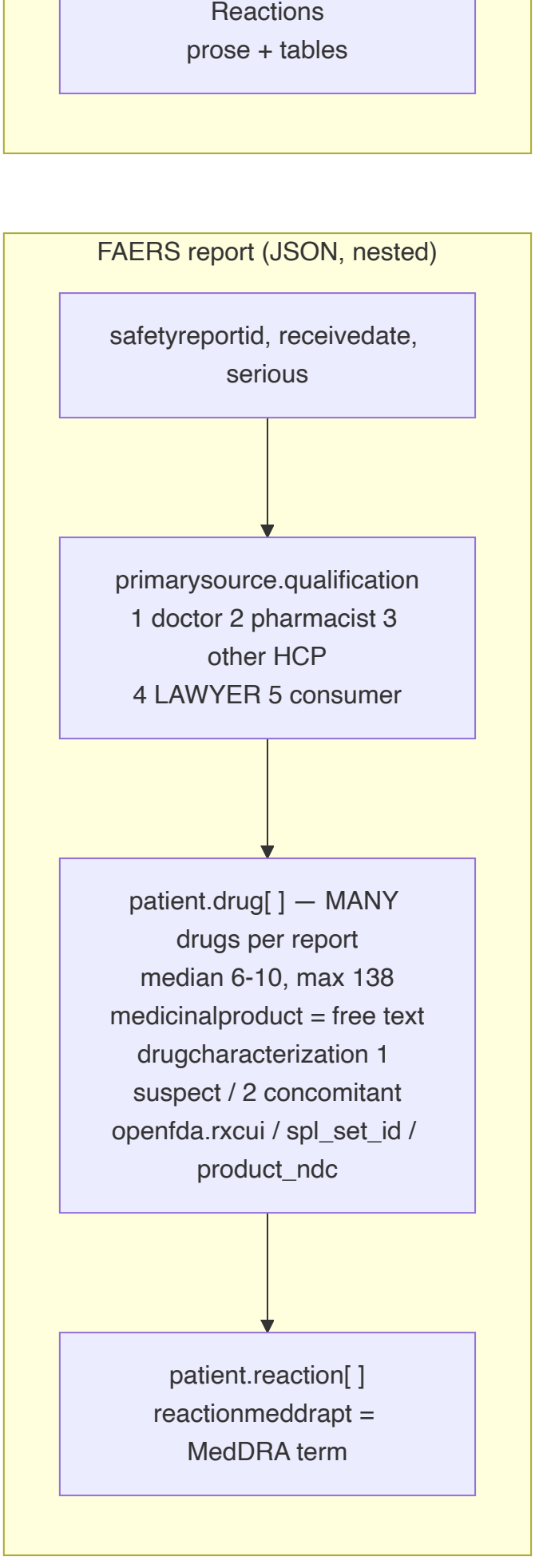
How each source is shaped (one record = one box)



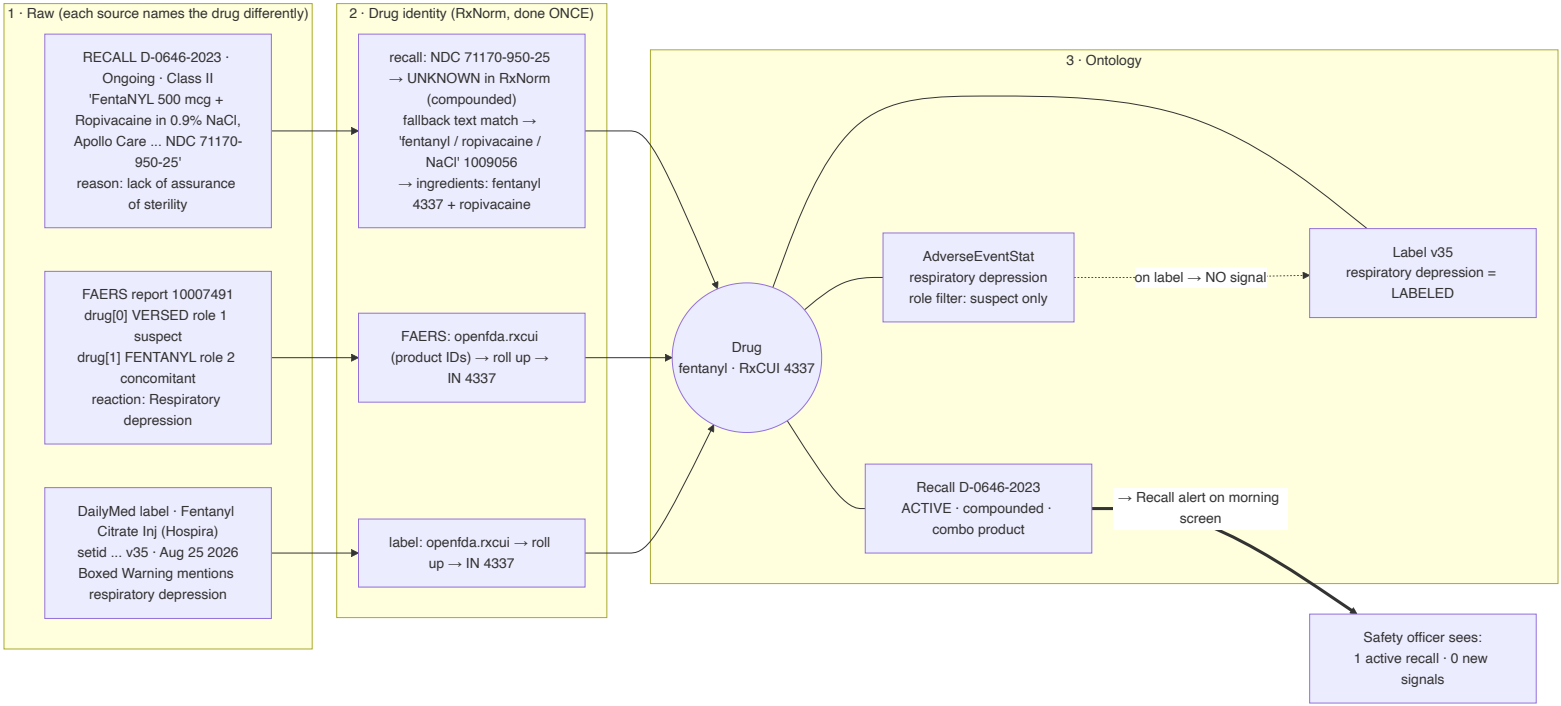




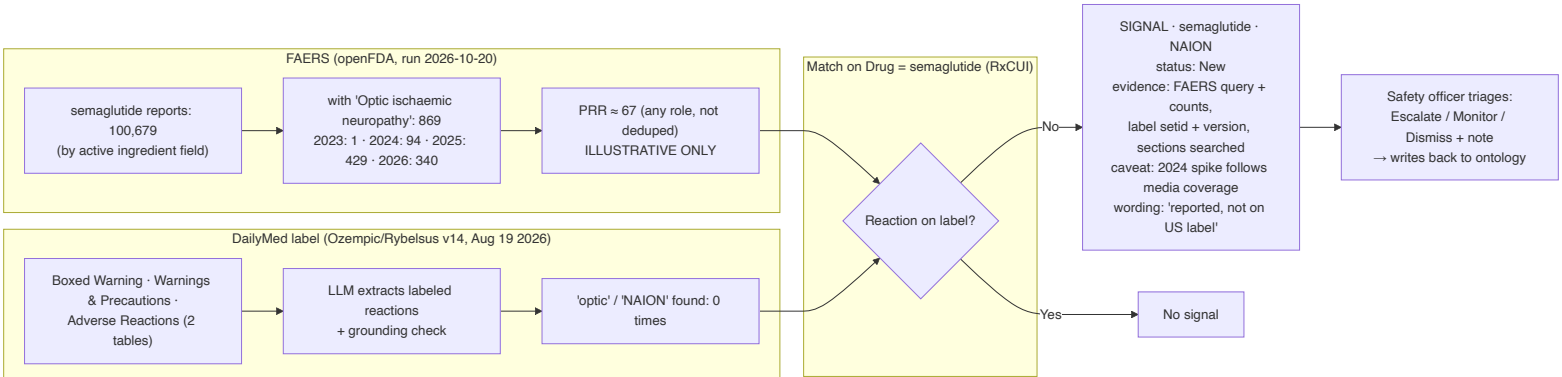


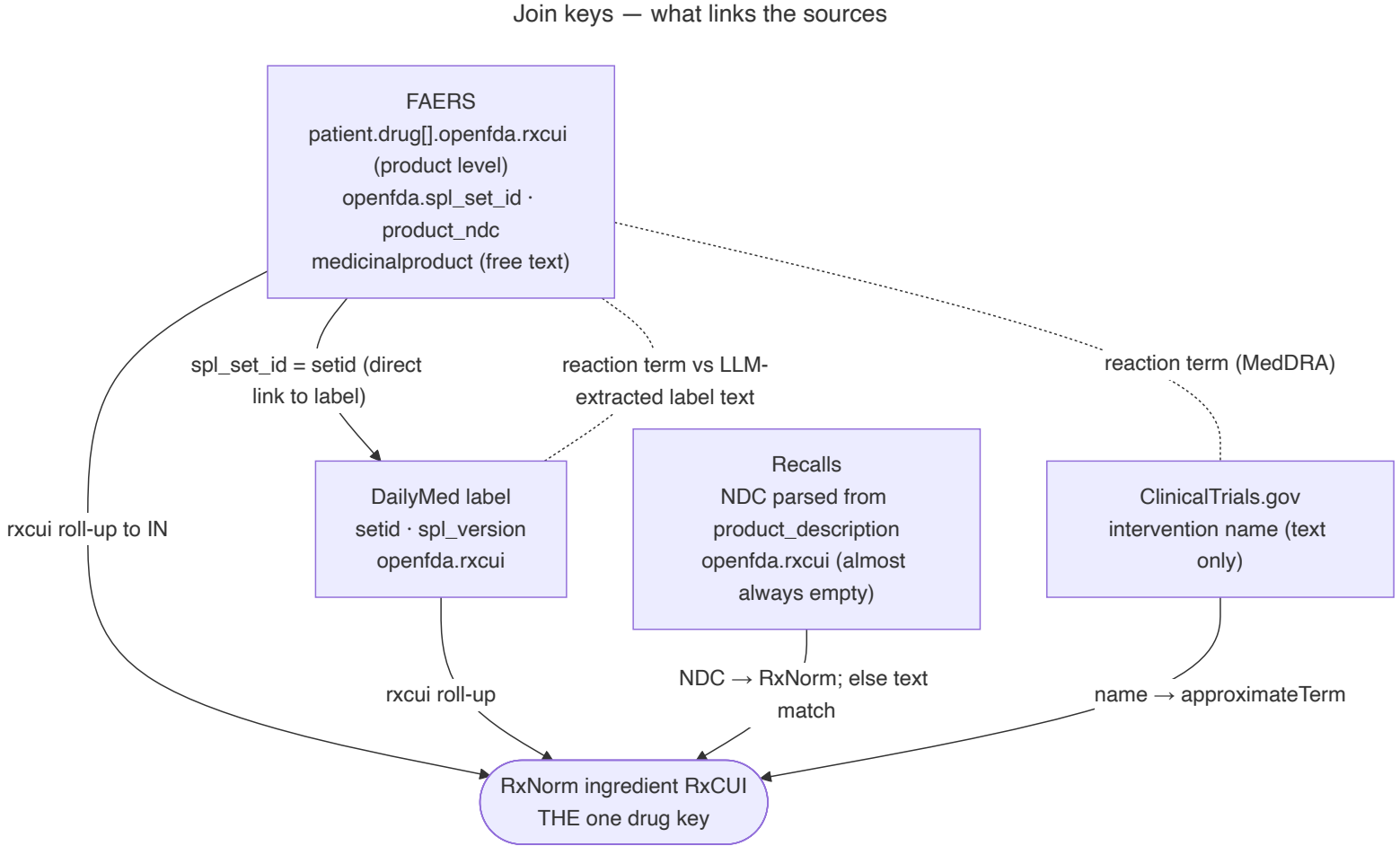


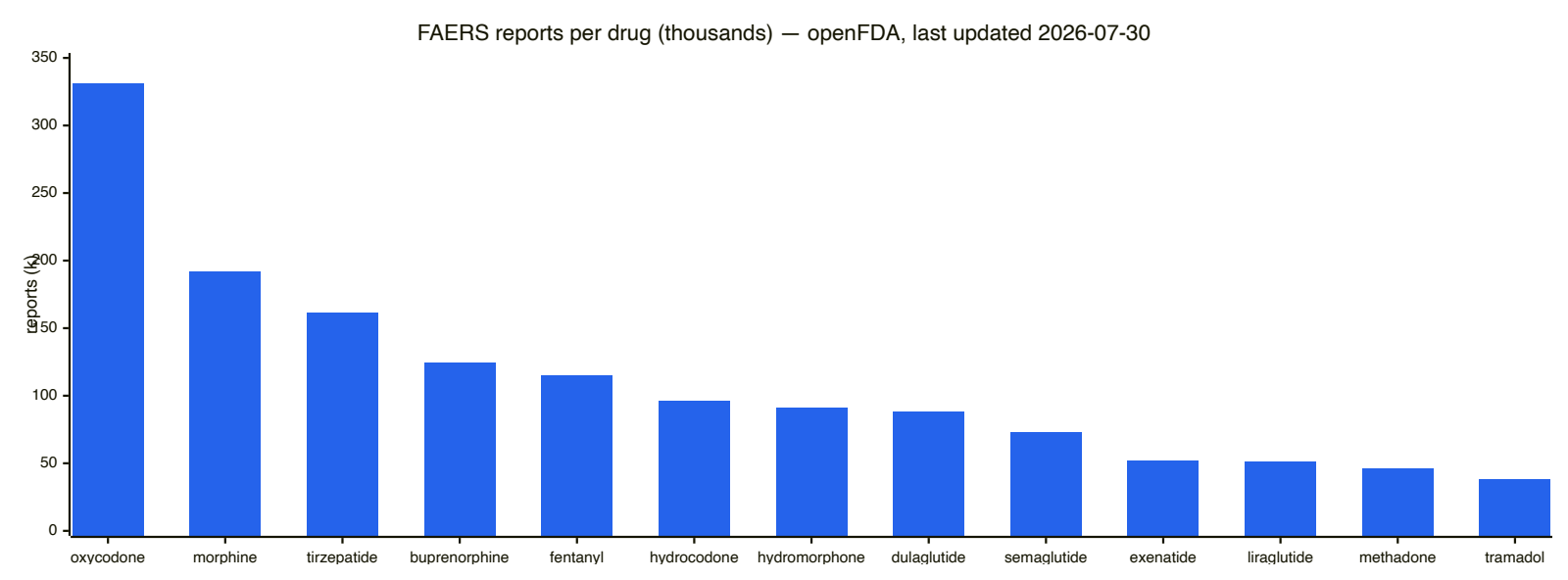
Example A — fentanyl: three sources, three shapes, one Drug

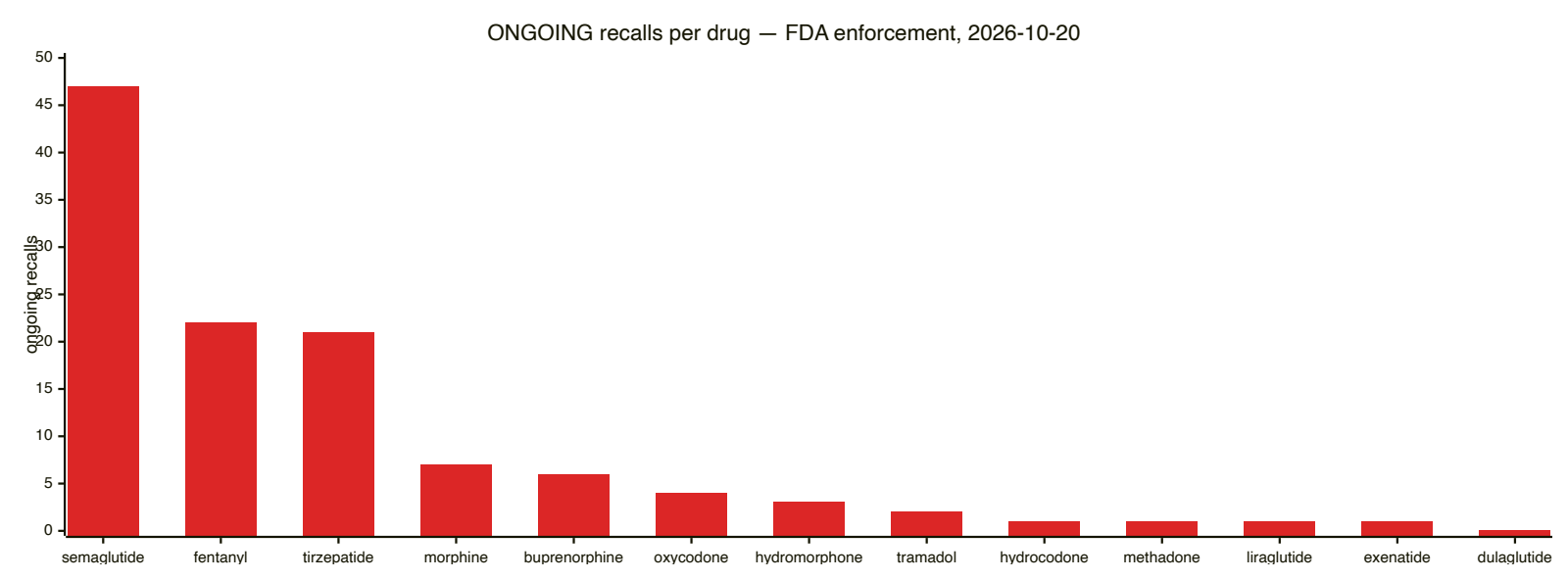


Example B — semaglutide + NAION: reported but not on the US label




Join keys — what links the sources
FAERS
patient.drug[].openfda.rxcui
(product level)
openfda.spl_set_id ·
product_ndc
medicinalproduct (free text)
spl_set_id = setid (direct
link to label)
reaction term vs LLM-
extracted label text
reaction term (MedDRA)
rxcui roll-up to IN
DailyMed label
setid · spl_version
openfda.rxcui
Recalls
NDC parsed from
product_description
openfda.rxcui (almost
always empty)
ClinicalTrials.gov
intervention name (text
only)
rxcui roll-up
NDC → RxNorm; else text
match
name → approximateTerm
RxNorm ingredient RxCUI
THE one drug key


FAERS reports per drug (thousands) — openFDA, last updated 2026-07-30
Reports (k)
350
300
250
200
150
100
50
0
oxycodone
morphine
tirzepatide
buprenorphine
fentanyl
hydrocodone
hydromorphone
dulaglutide
semaglutide
exenatide
liraglutide
methadone
tramadol


ONGOING recalls per drug — FDA enforcement, 2026-10-20
ongoing recalls
50
45
40
35
30
25
20
15
10
5
0
semaglutide
fentanyl
tirzepatide
morphine
buprenorphine
oxycodone
hydromorphone
tramadol
hydrocodone
methadone
liraglutide
exenatide
dulaglutide

One FAERS report → List 1 · One label → List 2 (Part B) · Compare → Signal

